

Atreya Rao

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Education

VIT Amravati

B.Tech in Computer Science June 2017 - June 2021

GPA: 8.73

Experience

Deloitte USI, Hyderabad - Software Engineer 2 Sept 2025 - Present

- Built a Node.js/NestJS application from scratch to migrate a business insurance client's policy-issuance and underwriting workload - supporting billions of dollars in insured business - from serverless to AWS EKS.
- Developed and deployed a React front-end application running on AWS EKS, giving the insurance client a unified, scalable policy-building platform.
- Built CI/CD pipelines using GitHub Actions, Kubernetes, and ArgoCD to automate build, test, and GitOps-based deployment to EKS, reducing manual release effort.
- Designed and implemented a disaster recovery (DR) strategy for the insurance platform on AWS EKS, including automated backups and multi-AZ failover to minimize downtime risk.
- Built and scaled infrastructure using Terraform (Infrastructure as Code), provisioning EKS clusters, networking, and supporting AWS resources across dev, staging, and production environments.
- Optimized application performance to reliably handle 1,000 requests per second (RPS) at 150 ms p95 latency, supporting 99.5% uptime for the insurance platform.
- Designed application architecture and coding standards to improve code quality and maintainability.
- Implemented a unit test strategy achieving 90% coverage, increasing reliability and easing refactoring.
- Authored internal onboarding wiki documentation and legacy serverless analysis to accelerate new team member ramp-up and guide migration momentum.
- Deployed and configured Istio service mesh (via Helm charts) to manage pod-to-pod traffic and enforce token-based authentication, strengthening the security posture of the microservices architecture.

EmployWise, Gurgaon (Remote) - Lead Developer Dec 2020 - Sept 2025

- Designed and developed scalable UI components using TypeScript, React.js, and MUI to enhance user experience.
- Successfully built and implemented a ticketing system with 17 features in just 6 months, resulting in 100% savings by eliminating the need for a third-party solution, using the MERN stack.
- Refactored and modernized React Native applications to support the latest package updates, improving code maintainability and performance.
- Developed 50+ complex pages with interactive dashboards and graphs, ensuring seamless data visualization for HR operations.
- Implemented lazy loading, memoization, and state management techniques (Redux, Context API) to optimize performance.
- Led technical interviews and successfully recruited 15+ frontend and backend engineers to scale the development team.
- Enhanced frontend security by implementing strict CSP policies, XSS protection, and input validation.
- Enhanced mobile app security by implementing certificate pinning, root detection, and SQL injection prevention, securing API endpoints with OAuth 2.0, JWT, etc.
- Migrated the React project to an Nx-based monorepo.
- Worked closely with UX/UI designers to iterate on designs based on user feedback and analytics-driven insights.

Skills

Frontend Development: JavaScript, React JS, Next JS, Redux, Nx Monorepo, TypeScript, Jest, HTML / CSS, Vite

Backend Development: Node JS, Nest JS, Microservices, Express JS, MongoDB, REST API, AI Agents, RAG, Langchain

Cloud & DevOps: AWS (EKS, Beanstalk, EC2, Route53, Amplify, S3, DynamoDB, CloudWatch), Kubernetes, ArgoCD, GitHub Actions, Docker

Mobile Development: React Native, Expo, Xcode, Android Studio

Projects

Ode To Paris [Freelancing] - *JavaScript, HTML/CSS, React JS, Next JS, Node JS, MongoDB*

Developed a full SaaS platform for a tour company in Paris, including a public-facing website for bookings and a robust admin dashboard for managing tours, payments, schedules, and customer data. Features integrated payments, email automation, SEO, and multi-guide support. Built with React, Node.js, MongoDB, and Next.js - scalable and designed for both solo guides and growing teams.

<https://www.odetoparis.com/>

We Residents [Personal] - *React JS, Node JS, MongoDB, Express*

Developed We Residents, an innovative property management platform designed to enhance residential living experiences by streamlining communication and fostering community engagement. Built and integrated a comprehensive suite of features to cater to the diverse needs of property administrators and residents - 14+ features and 50+ screens (web and mobile), all built independently.

<https://app.weresidents.com/>

Ticketing System [Work] - *React JS, MongoDB, Graph API (Microsoft), Node JS, Express JS*

Developed a robust ticketing system using the MERN stack as an alternative to Freshdesk, with key features including email notifications via Graph API, user roles such as agents and admins, and ticket properties like status and problem area. This system streamlined issue tracking and resolution, enhancing operational efficiency - completely built independently.

<https://support.myemploywise.com/>